- **1. Perform a simple calculation** (p. 1)
  - **Analysis:** Solving this problem requires basic arithmetic operations. Specifically, it involves calculating the difference in weight between the two eggs ($110\text{g} - 78\text{g} = 32\text{g}$), then dividing this difference by the initial weight of the first egg ($32\text{g}/78\text{g}$), and finally multiplying the result by 100 to get the percentage ($(32/78) \times 100\% \approx 41\%$). This is a direct, single-step or few-step calculation.
- **5. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms** (p. 1)
  - **Analysis:** Students need to understand the definition of "how many percent heavier" and know how to apply the corresponding percentage change formula: $\text{Percentage Change} = ((\text{New Value} - \text{Old Value})/\text{Old Value}) \times 100\%$. This requires recalling and applying known mathematical rules and algorithms.
- **7. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations** (p. 1)
  - **Analysis:** This ability involves extracting numerical information (78 g and 110 g) from the text and substituting them into the formula for numerical manipulation.
- **13. Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy** (p. 1)
  - **Analysis:** The problem is set in the context of penguin egg weights. While the context itself doesn't change the calculation method, understanding that it's a real-world problem about "weight" and "percentage" helps in comprehending the goal and judging the reasonableness of the final answer (e.g., the answer is a positive percentage because the second egg is heavier).
- **8. Articulate a solution, showing and/or summarising and presenting intermediate mathematical results**
  - **Analysis:** Although it's a multiple-choice question, the process of finding the correct answer involves intermediate results (weight difference of 32 g, ratio of 0.4102...). Clearly showing or mentally summarising these steps is crucial for reaching an accurate final result.

- **1. Perform a simple calculation** (1.3.2, 1.4.14)
  - **分析：** 解决此问题需要进行基本的算术运算。具体来说，需要计算两个蛋的重量差 ($110\text{g} - 78\text{g} = 32\text{g}$)，然后将这个差除以第一个蛋的原始重量 ($32\text{g}/78\text{g}$)，最后将结果乘以 100 得到百分比 ($(32/78) \times 100\% \approx 41\%$)。这是一个直接的、单一步骤或少数几个步骤的计算。
- **5. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms** (1.2.5, 1.4.12)
  - **分析：** 学生需要理解“百分比重多少”这一概念的定义，并知道如何应用相应的百分比变化公式：$\text{百分比变化} = ((\text{新值} - \text{旧值})/\text{旧值}) \times 100\%$。这需要调用并应用已知的数学规则和算法。
- **7. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations** (1.2.5)
  - **分析：** 这项能力涉及从文本中提取数值信息（78 g 和 110 g），并将它们代入公式中进行数值操作。
- **13. Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy** (1.2.6)
  - **分析：** 该问题以企鹅蛋的重量为背景，虽然背景本身不影响计算方法，但理解这是一个关于“重量”和“百分比”的现实世界问题有助于理解问题目标，并判断最终答案是否合理（例如，答案是一个正百分比，因为第二个蛋更重）。
- **8. Articulate a solution, showing and/or summarising and presenting intermediate mathematical results**
  - **分析：** 虽然题目是多项选择题，但找到正确答案的过程涉及中间结果（重量差 32 g、比值 0.4102...），清晰地展示或在脑海中总结这些步骤对于得出准确的最终结果至关重要。